

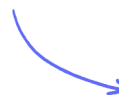
IGCSE • Cambridge (CIE) • Maths

 25 mins  4 questions

Modelling: River Bends

Modelling

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Total Marks

/25

- 1 (a)** This task looks at how to model the path of a bending river, using approximations to predict the speed of the river.

When viewed from above or drawn on a map, the path of a river is not a straight line but instead has many bends, especially as the river approaches the sea.

An S-bend is a bend in one direction followed immediately by a bend in the opposite direction. The map below shows a river path completing **two** identical S-bends before it reaches the sea (the water in the river flows from right to left).

The x -axis is the centre line of the river path, about which the river bends, measured inland from the sea. The y -axis is perpendicular to the x -axis. The units on both axes are kilometres.

The flat land either side of the river path is called the flood plain. The lines $y = 2.4$ and $y = -2.4$ indicate the boundaries of the flood plain.

Find the value of L from the diagram.

(1 mark)

(b) A geographer wants to model the river path shown in part (a).

They suggest two possible models, where angles are measured in degrees:

$$y = A \sin\left(\frac{360x}{L}\right)$$

$$y = A \cos\left(\frac{360x}{L}\right)$$

Decide which of the models is **not** suitable for the river shown in the diagram.

Give a reason for your answer.

(1 mark)

(c) Substitute the values of A and L into the **correct** model from part (b).

Simplify your answer.

(1 mark)

(d) Use the model in part (c) to find the perpendicular distance from the point on the centre line that is 500 metres inland from the sea, to the path of the river.

Give your answer correct to 3 significant figures.

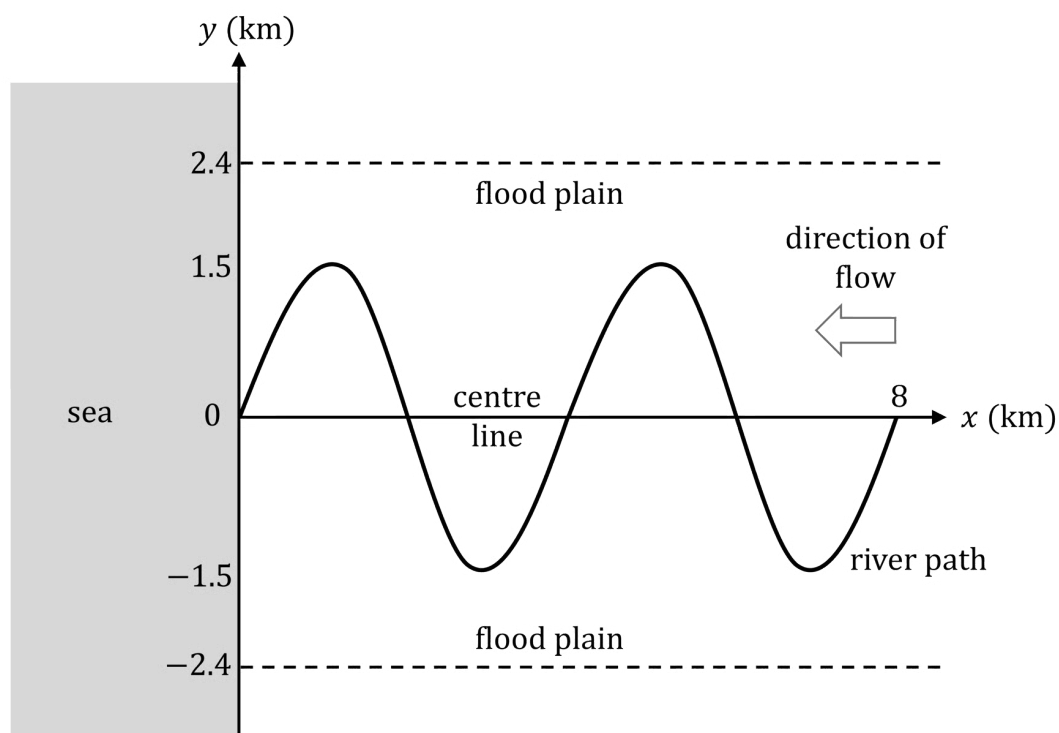
(2 marks)

- 2 (a)** A geographer is studying the path of a bending river. As shown on the diagram below, the river path completes two identical S-bends (a bend in one direction followed immediately by a bend in the opposite direction) before it reaches the sea.

The geographer models the path using the equation

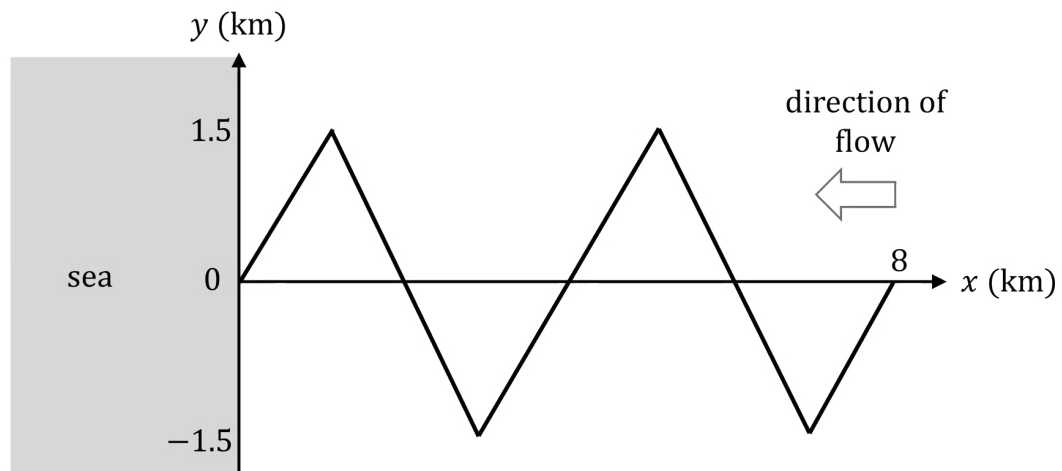
$$y = 1.5 \sin(90x)$$

where x is the distance along the centre line from the sea, and y is the perpendicular distance of the river from the centre line. Both x and y are measured in kilometres.



The geographer wants to find the length of the river using the model, but does not know how to measure the length of a trigonometric curve.

Instead, they decide to approximate the length of the river by dividing it into straight-line segments, as shown.



Each straight line segment passes through the correct maximum point, minimum point and x -intercept.

Show that the river is approximately 14.4 km in length.

Explain your method.

(2 marks)

(b) The geographer wants to estimate the speed of the river.

They place a paper boat into the river at $x = 8$ and time how long it takes for the paper boat to reach the sea, at $x = 0$.

The total time taken is 2 hours and 24 minutes.

Use this information, and the rounded value of the length in part (a), to estimate the speed of the river.

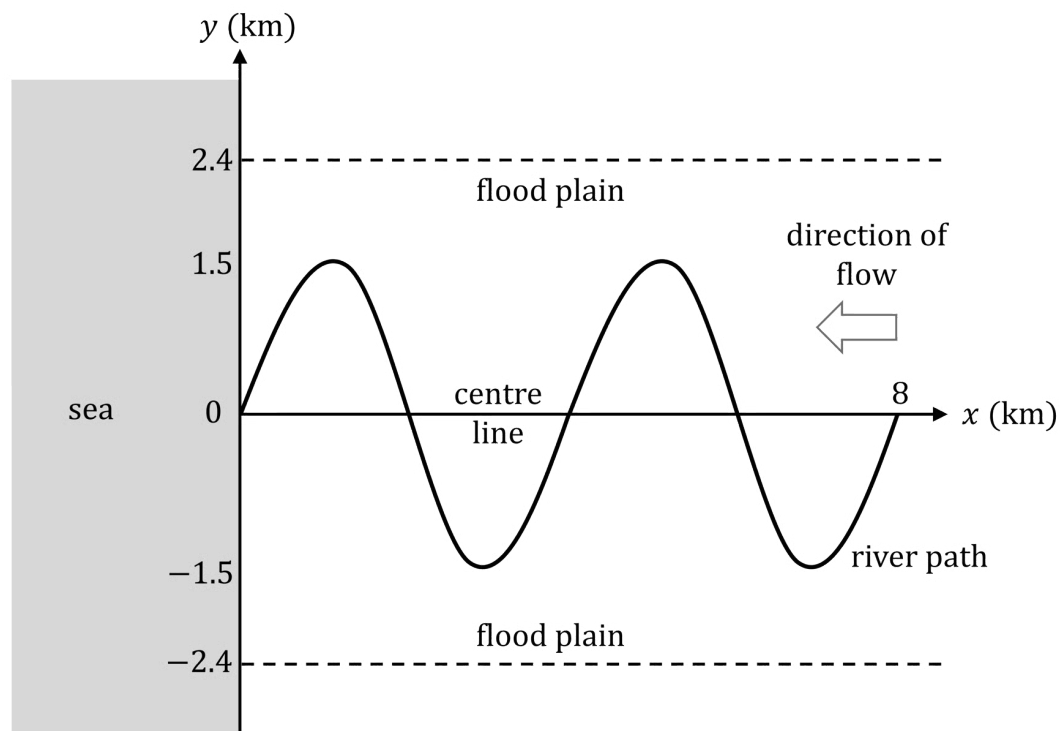
Give your answer in km/h.

(2 marks)

- (c)** Based on the approximation made in part (a), explain whether the true speed of the river is likely to be higher or lower than the value calculated in part (b).

(2 marks)

- 3 (a)** A geographer is studying the path of a bending river. As shown on the diagram below, the river path completes two identical S-bends (a bend in one direction followed immediately by a bend in the opposite direction) before it reaches the sea.



The geographer used a straight-line approximation to approximate the length of the river between $x = 8$ and $x = 0$ as 14.4 km.

Originally the geographer assumed that the speed of the river was constant. But in reality, the speed of the river increases by 2 km/h as it approaches the sea.

The geographer creates a simple model to show this change:

- The S-bend from $x = 8$ to $x = 4$ has a velocity of v km/h
- The S-bend from $x = 4$ to $x = 0$ has a velocity of $(v + 2)$ km/h

The approximate length of the river is rounded to the **nearest integer**.

Find, in terms of v , a formula for the approximate time taken (in hours) to travel along the S-bend from $x = 8$ to $x = 4$.

(1 mark)

- (b) Find, in terms of v , a formula for the approximate time taken (in hours) to travel along the S-bend from $x = 4$ to $x = 0$.

(1 mark)

- (c) Show that the approximate total time (in hours) to travel along both S-bends is given by the formula

$$\frac{14(v+1)}{v(v+2)}$$

(2 marks)

- (d) The geographer measured that the total time taken by an object placed in the river at $x = 8$ to reach the sea at $x = 0$ is 2 hours and 24 minutes.

Use this total time to show that v satisfies the equation

$$6v^2 - 23v - 35 = 0$$

(2 marks)

- (e) Solve the equation in part (d) to find v .

(2 marks)

- 4 (a)** A geographer is studying the path of a bending river. The river path completes a number of S-bends (a bend in one direction followed immediately by a bend in the opposite direction) before it reaches the sea.

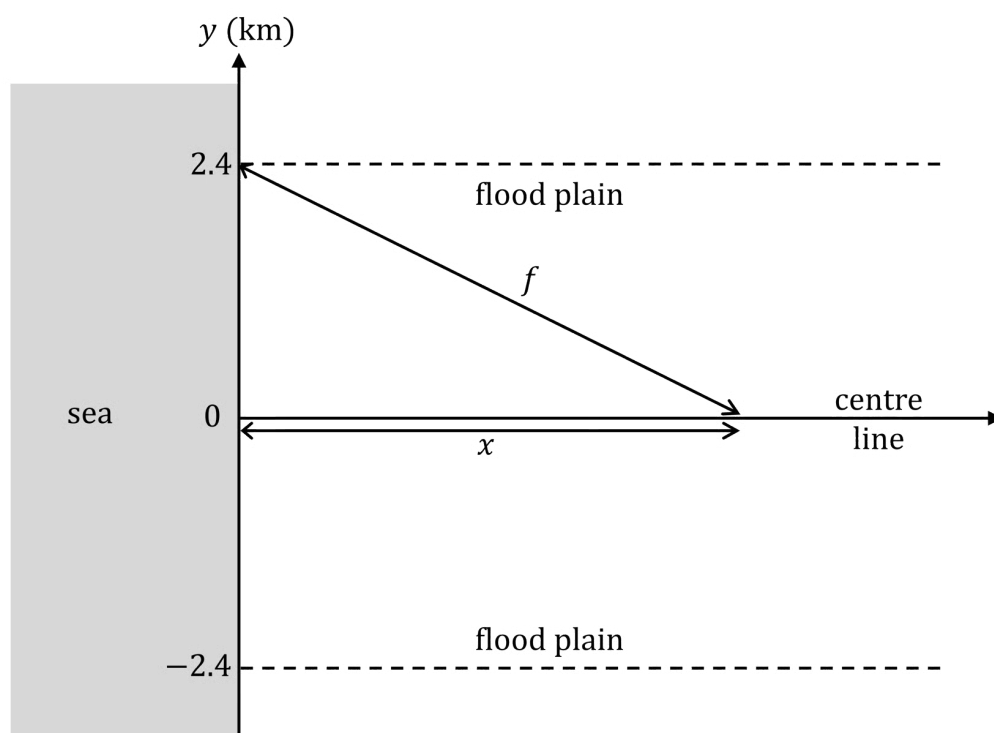
The geographer would like to use a model for a river path approaching the sea that has the form

$$y = A \sin(90x)$$

where x is the perpendicular distance from the sea, and y is the perpendicular distance of the river from the centre line where $y = 0$. Both x and y are measured in kilometres.

The amplitude, A km, of an S-bend is the maximum distance (in the y -direction) from the centre line to the river path.

In this model, the amplitude of an S-bend, A , at a distance x km inland from the sea is inversely proportional to f , the distance to where the flood plain boundary meets the sea, shown below:



Show that this model has the equation

$$y = \frac{k}{\sqrt{2.4^2 + x^2}} \sin(90x)$$

where k is an unknown constant of proportionality.

(2 marks)

- (b)** In this model, it is known that $y = 1.5$ when $x = 1$.

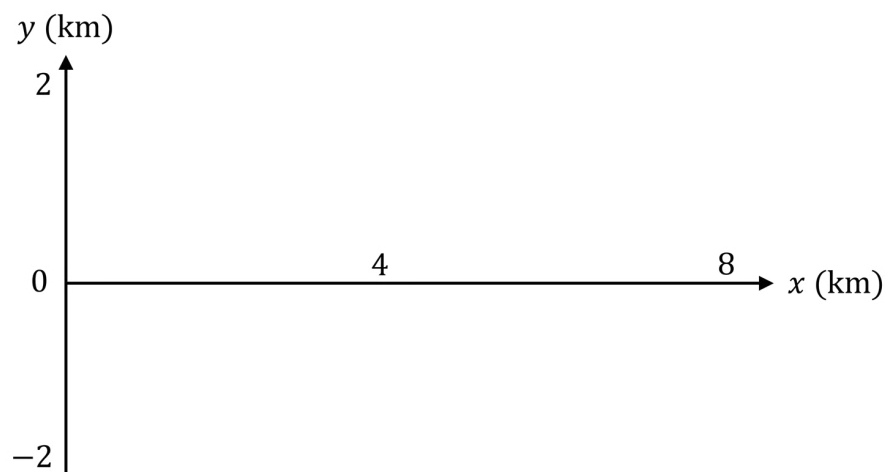
Use this information to show clearly that

$$k = 3.9$$

(1 mark)

- (c)** Sketch the model on the axes provided.

Label any x -intercepts clearly.



(2 marks)

- (d)** Describe how the S-bends change as the river approaches the sea.

(1 mark)